

FASTAPI & JWT AUTHENTICATION SYSTEM



Technical Concepts with Real-Life Analogies

FastAPI / JWT Term	Technical Meaning	Real-Life Analogy
 Authentication	Verifying the user's identity	 Showing your ID card at the office gate
 Authorization	Checking what the user is allowed to access	 Security checking which rooms you can enter
 JWT Token	Digitally signed token used for user verification	 Employee ID Card
 Payload	Data stored inside the token	 Details printed on the ID Card (Name, ID, Role)
 Secret Key	Secret string used to sign and verify tokens	 Company's official stamp or seal
 Algorithm (HS256)	Method used to create the digital signature	 Procedure used to create the official stamp
 Signature	Encrypted verification part of JWT	 Stamp impression on the ID Card
 Header	Metadata about the token	 Card type information (Employee Card, Visitor Card)
 Encode	Convert payload into a JWT token	 Printing and stamping the ID Card
 Decode	Read and verify the token	 Inspecting the ID Card and stamp
 Access Token	Token used to access protected APIs	 Entry pass shown at every security checkpoint
 Protected API	Endpoint requiring authentication	 Restricted office area
 Login	User submits credentials	 Employee shows username and password
 Username	User identifier	 Employee Name
 Password	Secret credential used for login	 Personal PIN or Password
 Token Expiry	Time limit after which token becomes invalid	 ID Card validity period
 Bearer Token	Token sent in request headers	 Carrying your ID Card while entering departments
 <code>create_token()</code>	Function that generates a JWT token	 HR issuing a new employee ID Card
 <code>jwt.encode()</code>	Creates a signed JWT token	 Printing and stamping the ID Card
 <code>jwt.decode()</code>	Verifies and reads JWT token	 Security verifying the ID Card
 User Database	Stores user credentials	 Employee records maintained by HR
 Role (Admin/User)	Determines access level	 Manager vs Employee access privileges